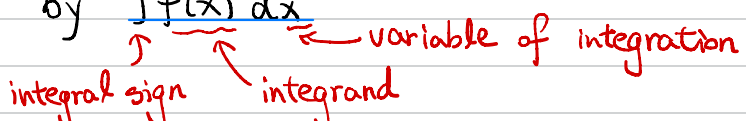


Antiderivatives & Indefinite Integrals

Antiderivatives: A function F is an antiderivative of f on an interval I if $F'(x) = f(x)$ for all x in I .

Let $F(x)$ be a function such that $F'(x) = f(x)$, then the most general antiderivative of f on I is $F(x) + C$, where C is an arbitrary constant.

Indefinite Integrals: The collection of all antiderivatives of f is called the indefinite integral of f with respect to x , and it is denoted by $\int f(x) dx$.



Examples: Let $f(x) = 2x$, and let C be an arbitrary constant. Which of the following statements is FALSE?

- A. $x^2 + 1$ is an antiderivative of $f(x)$
- B. x^2 is an antiderivative of $f(x)$
- ☒ C. x^2 is the indefinite integral of $f(x)$
- D. $x^2 + C$ is the indefinite integral of $f(x)$

(Riemann Sum)

Area and Estimating with Finite Sums

area below a positive & continuous function on $[a, b]$

Let $A = \text{area of } R$

Upper sum for A

$$= 1 \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2}$$

$$\text{or} \\ = 1 \cdot \frac{1}{4} + \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4}$$

Lower sum for A

$$= \frac{3}{4} \cdot \frac{1}{2} + 0 \cdot \frac{1}{2}$$

$$\text{or} \\ = \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4} + 0 \cdot \frac{1}{4}$$

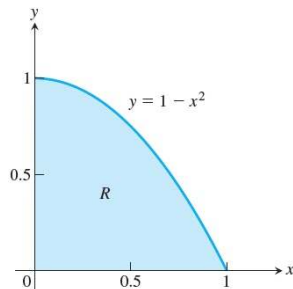


FIGURE 5.1 The area of a region R cannot be found by a simple formula.

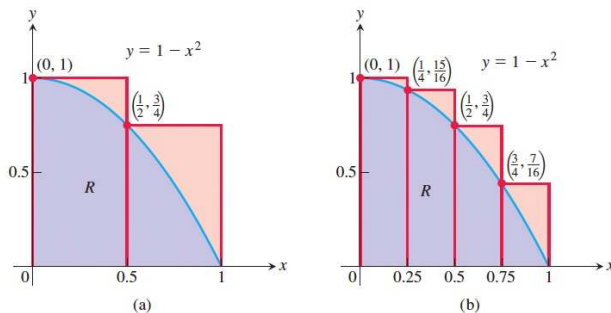


FIGURE 5.2 (a) We get an upper estimate of the area of R by using two rectangles containing R . (b) Four rectangles give a better upper estimate. Both estimates overshoot the true value for the area by the amount shaded in light red.

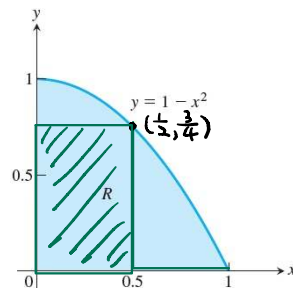


FIGURE 5.1 The area of a region R cannot be found by a simple formula.

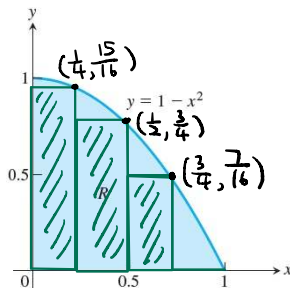


FIGURE 5.1 The area of a region R cannot be found by a simple formula.

Finite Sum

$$\text{e.g. } \sum_{i=1}^n \left(\frac{i^2}{n} + \frac{2i}{n^2} + 1 \right) = \frac{1}{n} \sum_{i=1}^n i^2 + \frac{2}{n^2} \sum_{i=1}^n i + \sum_{i=1}^n 1$$

$$\sum_{i=1}^n 1 = n$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^3 = \left(\frac{n(n+1)}{2} \right)^2$$

A Formula for the Riemann Sum with Equal Width Subintervals

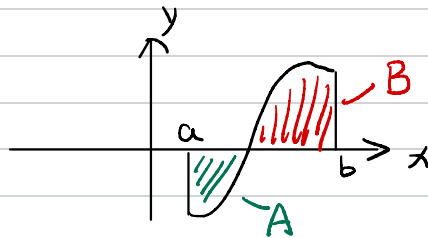
$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{i=1}^n f\left(a + i\left(\frac{b-a}{n}\right)\right) \left(\frac{b-a}{n}\right)$$

$$\text{average value} = \frac{1}{b-a} \int_a^b f(x) dx$$

Geometric Meaning of the Definite Integral

Net area below the graph of $f(x)$

$$\text{e.g. } \int_a^b f(x) dx = B - A$$



The Fundamental Theorem of Calculus (FTC)

Let f be a function that is continuous on the interval $[a, b]$, and let F be any antiderivative of f . Then:

$$\int_a^b f(x) dx = F(x) \Big|_a^b = F(b) - F(a)$$

Average value:
(if f is cts on $[a, b]$)

$$\frac{\text{Area}}{b-a} = \frac{\int_a^b f(x) dx}{b-a} = \frac{F(b) - F(a)}{b-a}$$

2nd FTC If f is cts on the interval $[a, b]$.

$$\frac{d}{dx} \left[\int_{a(x)}^{b(x)} f(t) dt \right] = f(b(x)) b'(x) - f(a(x)) a'(x)$$

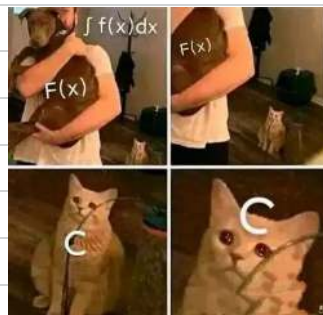
In particular $\frac{d}{dx} \left[\int_c^{b(x)} f(t) dt \right] = f(b(x)) b'(x)$

when c is a constant.

Integration by Substitution (u-sub)

Q: Which kind of integrands can the u-sub method be applied to?

A: When $f(x) f'(x)$ is included



Steps to Perform U-substitution

① Find $u=f(x)$

② Using dx to express du , i.e. $dx = \frac{1}{f'(x)} du$
b/c $\frac{du}{dx} = f'(x) \Rightarrow dx = \frac{1}{f'(x)} du$

③ Plugging ① & ② into the original question:

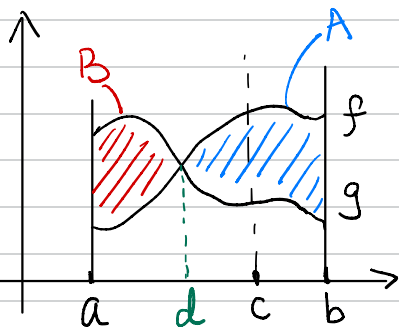
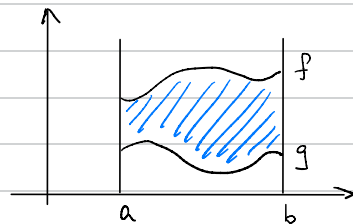
$$\int f(x) \cdot f'(x) dx = \int u \cdot \cancel{f'(x)} \left(\frac{1}{\cancel{f'(x)}} \right) du = \int u du$$

④ Evaluating & Plugging $u=f(x)$ back (Don't forget $+C$!)

Area Between Curves

If f, g are continuous w/ $f(x) \geq g(x)$ throughout $[a, b]$, then the area of the region between the curves $y=f(x)$ and $y=g(x)$ from a to b is the integral

$$A = \int_a^b [f(x) - g(x)] dx$$



$$\int_a^b (f(x) - g(x)) dx = A - B \neq \text{Area between curves}$$

$$\text{Area between curves} = A + B$$

Area between curves

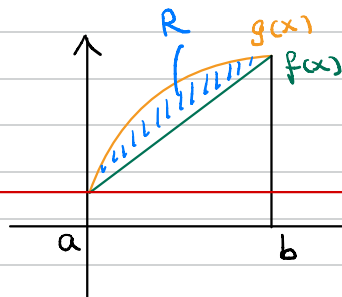
$$= \underbrace{\int_a^d (g(x) - f(x)) dx}_B + \underbrace{\int_d^b (f(x) - g(x)) dx}_A$$

Volumes by the Disk Method

$f(x)$ revolved about x -axis

$$V = \pi \int_a^b (f(x) - 0)^2 dx$$

$\leftarrow$ b/c x -axis is $y=0$



region R revolved about $y=1$

$\leftarrow$ $g(x)$ first b/c $g(x)$ is farther from $y=1$

$$V = \pi \int_a^b (\underline{g(x)} - 1)^2 - (\overline{f(x)} - 1)^2 dx$$

$\leftarrow$ b/c revolved about $y=1$

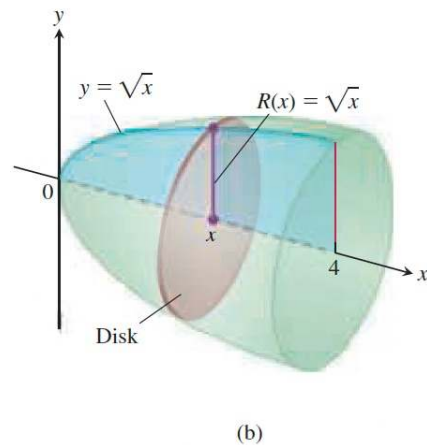
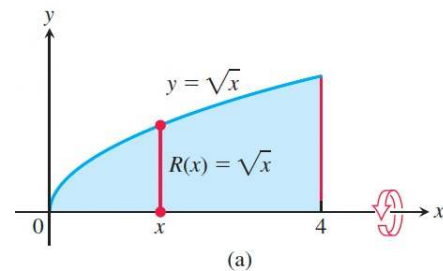


FIGURE 6.8 The region (a) and solid of revolution (b) in Example 4.

Volumes by the Shell Method

$$V = \int_a^b 2\pi (\text{shell radius})(\text{shell height}) dx$$

Steps to think about:

① What is the interval of integration?
 $a = ?$ $b = ?$

② What is the shell radius?

③ What is the shell height?

If you don't know where to start, it is always helpful to first try to draw the cylindrical shell.

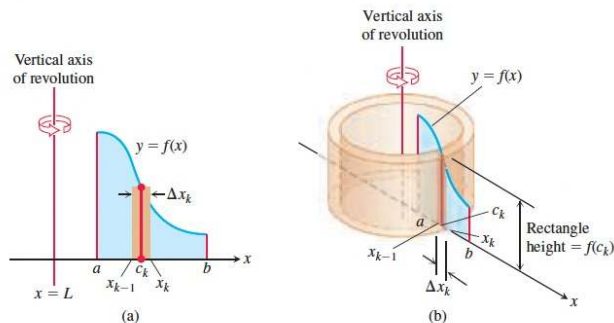


FIGURE 6.19 When the region shown in (a) is revolved about the vertical line $x = L$, a solid is produced which can be sliced into cylindrical shells. A typical shell is shown in (b).

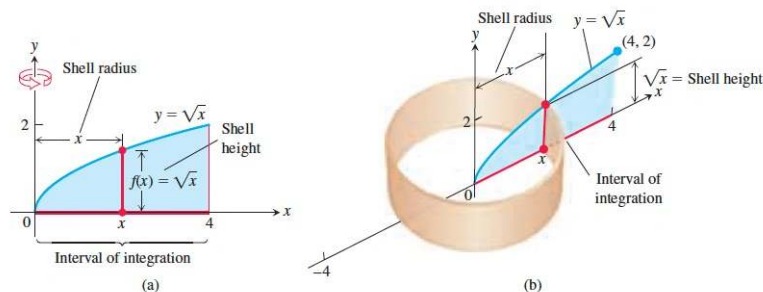


FIGURE 6.20 (a) The region, shell dimensions, and interval of integration in Example 2. (b) The shell swept out by the vertical segment in part (a) with a width Δx .

Integration by Parts

When we have $\int f(x)g(x)dx$ and u-sub doesn't work.

Goal: Convert $f(x)g(x)$ into one expression $h(x)$

Formula: $\int u dv = uv - \int v du$ (*)

Recipe: ① Figure out what are u and dv

e.g. $u = f(x)$ $dv = g(x)dx$

Normally: $f(x)$ is easier to differentiate
 $g(x)$ is easier to integrate

② Compute du and v
e.g. $du = f'(x)dx$ $v = \int g(x)dx$

③ Plug in all these info (u, v, du, dv) into (*)

Powers and Products of Trig Functions

TRIG IDENTITIES

Reciprocal Identities

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Quotient Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$1 = \sin^2 \theta + \cos^2 \theta$$

$$\sec^2 \theta = \tan^2 \theta + 1$$

$$\csc^2 \theta = 1 + \cot^2 \theta$$

Power Reducing Identities

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

DERIVATIVES

Trig Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

INTEGRALS

Trig Functions

$$\int \sin u \, du = -\cos u + c$$

$$\int \sec u \, du = \ln|\sec u + \tan u| + c$$

$$\int \sec^2 u \, du = \tan u + c$$

$$\int \cos u \, du = \sin u + c$$

$$\int \csc u \, du = \ln|\csc u - \cot u| + c$$

$$\int \csc^2 u \, du = -\cot u + c$$

$$\int \tan u \, du = \ln|\sec u| + c$$

$$\int \cot u \, du = \ln|\sin u| + c$$

Trigonometric Substitutions

TRIG IDENTITIES

Reciprocal Identities

$$\sin \theta = \frac{1}{\csc \theta}$$
$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$
$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$
$$\cot \theta = \frac{1}{\tan \theta}$$

Quotient Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$1 = \sin^2 \theta + \cos^2 \theta$$

$$\sec^2 \theta = \tan^2 \theta + 1$$

$$\csc^2 \theta = 1 + \cot^2 \theta$$

$$\sin^2(\theta) = 1 - \cos^2(\theta)$$
$$\cos^2(\theta) = 1 - \sin^2(\theta)$$

trig sub:

$$x = \tan(\theta)$$

$$\text{trig sub: } x = \cos(\theta) \Rightarrow \sin^2(\theta) = 1 - x^2 \Rightarrow \sec^2(\theta) = x^2 + 1$$
$$x = \sin(\theta) \Rightarrow \cos^2(\theta) = 1 - x^2$$

DERIVATIVES

Trig Functions

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

INTEGRALS

Trig Functions

$$\int \sin u \, du = -\cos u + c$$

$$\int \cos u \, du = \sin u + c$$

$$\int \tan u \, du = \ln|\sec u| + c$$

$$\int \sec u \, du = \ln|\sec u + \tan u| + c$$

$$\int \csc u \, du = \ln|\csc u - \cot u| + c$$

$$\int \cot u \, du = \ln|\sin u| + c$$

$$\int \sec^2 u \, du = \tan u + c$$

$$\int \csc^2 u \, du = -\cot u + c$$

Trigonometric Substitutions

$$a^2 - x^2 \Rightarrow x = \sin(\theta) \text{ or } x = \cos(\theta)$$

$$x^2 + a^2 \Rightarrow x = \tan(\theta)$$

$$x^2 - a^2 \Rightarrow x = \sec(\theta)$$

TRIG IDENTITIES

Reciprocal Identities

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Quotient Identities

$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$\cot \theta = \frac{\cos \theta}{\sin \theta}$$

Pythagorean Identities

$$1 = \sin^2 \theta + \cos^2 \theta$$

$$\sec^2 \theta = \tan^2 \theta + 1$$

$$\csc^2 \theta = 1 + \cot^2 \theta$$

Double Angle Identities

$$\sin 2\theta = 2 \sin \theta \cos \theta$$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\cos 2\theta = 1 - 2 \sin^2 \theta$$

Power Reducing Identities

$$\sin^2 \theta = \frac{1 - \cos 2\theta}{2}$$

$$\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$$

$$\tan^2 \theta = \frac{1 - \cos 2\theta}{1 + \cos 2\theta}$$

Partial Fractions

$$\int \frac{f(x)}{g(x)} dx$$

- the degree of $f(x)$ is less than the degree of $g(x)$
- Need to factorize $g(x)$

Some examples:

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{(x-a)} + \frac{B}{(x-b)}, \quad a \neq b$$

$$\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)^2} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}$$

Partial Fractions

$$\int \frac{f(x)}{g(x)} dx$$

- the degree of $f(x)$ is less than the degree of $g(x)$
- Need to factorize $g(x)$

Some examples:

$$\frac{px+q}{(x-a)(x-b)} = \frac{A}{(x-a)} + \frac{B}{(x-b)}, \quad a \neq b$$

$$\frac{px+q}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)(x-c)} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{x-c}$$

$$\frac{px^2+qx+r}{(x-a)(x-b)^2} = \frac{A}{x-a} + \frac{B}{x-b} + \frac{C}{(x-b)^2}$$

$$\frac{px^2+qx+r}{(x-a)(x^2+bx+c)} = \frac{A}{x-a} + \frac{Bx+C}{x^2+bx+c}$$

↑
This is an irreducible quadratic

L'Hôpital's Rule

Can be transformed
into $\frac{0}{0}$ or $\frac{\infty}{\infty}$, then
use L'Hôpital's Rule

Indeterminate Forms: $\frac{0}{0}, \frac{\infty}{\infty}, \infty \cdot 0, \infty - \infty$

L'Hôpital's Rule
can be applied
directly

Theorem: $f(a) = g(a) = 0$, and f and g are differentiable on an open interval I containing a , and that $g' \neq 0$ on I if $x \neq a$. Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

assuming that the limit on the right side of this equation exists.

Indeterminate Powers: $1^\infty, 0^0, \infty^0$

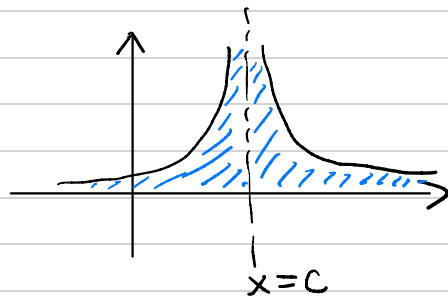
If $\lim_{x \rightarrow a} [\ln(f(x))] = L$, then

$$\lim_{x \rightarrow a} [f(x)] = \lim_{x \rightarrow a} [e^{\ln(f(x))}] = e^L$$

Improper Integrals Extension of definite Integrals

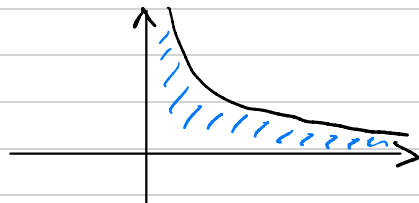
Case 1: Vertical Asymptote:

$$\int_a^b f(x) dx = \lim_{d \rightarrow c^-} \left[\int_a^d f(x) dx \right] + \lim_{g \rightarrow c^+} \left[\int_g^b f(x) dx \right]$$



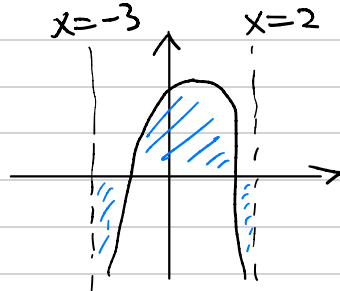
Case 2(a) Infinite Limits of integration.

$$\int_0^{\infty} f(x) dx = \lim_{c \rightarrow \infty} \left[\int_0^c f(x) dx \right]$$



Case 2(b)

$$\int_{-\infty}^{\infty} f(x) dx = \lim_{c \rightarrow -\infty} \left[\int_c^5 f(x) dx \right] + \lim_{d \rightarrow \infty} \left[\int_5^d f(x) dx \right]$$



Case 3: Two Vertical Asymptote

$$\int_{-3}^2 f(x) dx = \lim_{c \rightarrow 3^+} \left[\int_c^0 f(x) dx \right] + \lim_{d \rightarrow 2^-} \left[\int_0^d f(x) dx \right]$$

If any of the limit gives you infinity, the integral **diverges**

Infinite Sequences

An expression about n

$$\{a_n\} = \{a_1, a_2, \dots\}$$

starts from $n=1$

Monotonic:

- ① increasing
- ② decreasing
- ③ constant

lower bound (L):
for all n , $L \leq a_n$.

upper bound (U):
for all n , $U \geq a_n$

greatest lower bound (g.l.b.):
for all lower bounds L , $\text{g.l.b.} \geq L$

least upper bound (l.u.b.):
for all upper bounds U , $\text{l.u.b.} \leq U$.

DEFINITIONS The sequence $\{a_n\}$ **converges** to the number L if for every positive number ε there corresponds an integer N such that

$$|a_n - L| < \varepsilon \quad \text{whenever} \quad n > N.$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.

If $\{a_n\}$ converges to L , we write $\lim_{n \rightarrow \infty} a_n = L$, or simply $a_n \rightarrow L$, and call L the **limit** of the sequence (Figure 10.2).

DEFINITION The sequence $\{a_n\}$ **diverges to infinity** if for every number M there is an integer N such that for all n larger than N , $a_n > M$. If this condition holds we write

$$\lim_{n \rightarrow \infty} a_n = \infty \quad \text{or} \quad a_n \rightarrow \infty.$$

Similarly, if for every number m there is an integer N such that for all $n > N$ we have $a_n < m$, then we say $\{a_n\}$ **diverges to negative infinity** and write

$$\lim_{n \rightarrow \infty} a_n = -\infty \quad \text{or} \quad a_n \rightarrow -\infty.$$

Infinite Series : the sum of an infinite sequence of numbers

DEFINITIONS Given a sequence of numbers $\{a_n\}$, an expression of the form

$$a_1 + a_2 + a_3 + \cdots + a_n + \cdots$$

is an **infinite series**. The number a_n is the **n th term** of the series. The sequence $\{s_n\}$ defined by

$$s_1 = a_1$$

$$s_2 = a_1 + a_2$$

$$\vdots$$

$$s_n = a_1 + a_2 + \cdots + a_n = \sum_{k=1}^n a_k$$

$$\vdots$$

is the **sequence of partial sums** of the series, the number s_n being the **n th partial sum**. If the sequence of partial sums converges to a limit L , we say that the series **converges** and that its **sum** is L . In this case, we also write

$$a_1 + a_2 + \cdots + a_n + \cdots = \sum_{n=1}^{\infty} a_n = L.$$

If the sequence of partial sums of the series does not converge, we say that the series **diverges**.

If $|r| < 1$, the geometric series $a + ar + ar^2 + \cdots + ar^{n-1} + \cdots$ converges to $a/(1-r)$:

$$\sum_{n=1}^{\infty} ar^{n-1} = \frac{a}{1-r}, \quad |r| < 1.$$

If $|r| \geq 1$, the series diverges.

THEOREM 7 If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$.

The n th-Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \rightarrow \infty} a_n$ fails to exist or is different from zero.

THEOREM 8 If $\sum a_n = A$ and $\sum b_n = B$ are convergent series, then

- Sum Rule:** $\sum(a_n + b_n) = \sum a_n + \sum b_n = A + B$
- Difference Rule:** $\sum(a_n - b_n) = \sum a_n - \sum b_n = A - B$
- Constant Multiple Rule:** $\sum ka_n = k \sum a_n = kA$ (any number k).

THEOREM 7 If $\sum_{n=1}^{\infty} a_n$ converges, then $a_n \rightarrow 0$.

The n th-Term Test for Divergence

$\sum_{n=1}^{\infty} a_n$ diverges if $\lim_{n \rightarrow \infty} a_n$ fails to exist or is different from zero.

$\sum_{n=1}^{\infty} a_n$ converges $\Rightarrow a_n \rightarrow 0$ 

$a_n \rightarrow L \neq 0 \Rightarrow \sum_{n=1}^{\infty} a_n$ diverges

3. Determine if each statement below is always true or sometimes false. If the statement is false, provide a counterexample.

(a) If $\lim_{n \rightarrow \infty} a_n = 0$, then the series $\sum a_n$ converges.

False counterexample: $\sum_{k=1}^{\infty} \left(\frac{1}{k}\right)$

Harmonic Series

(b) If $\lim_{n \rightarrow \infty} a_n = 0$, then the sequence $\{a_n\}$ converges. True

(c) The series $\sum_{n=0}^{\infty} \frac{1}{n+1}$ diverges. True $\sum_{n=0}^{\infty} \left(\frac{1}{n+1}\right) = \sum_{n=1}^{\infty} \left(\frac{1}{n}\right)$

(d) The sum of two divergent series also diverges. False counterexample: $\sum_{k=0}^{\infty} [(-1)^k + (-1)^{k+1}]$

(e) The series $\sum_{n=1}^{\infty} r^n$ converges to $\frac{1}{1-r}$ if $|r| < 1$. False $\sum_{n=0}^{\infty} (r^n) = \frac{1}{1-r}$

$$\sum_{n=1}^{\infty} (r^n) = \frac{1}{1-r} - 1$$

The Integral Test

THEOREM 9—The Integral Test

Let $\{a_n\}$ be a sequence of positive terms. Suppose that $a_n = f(n)$, where f is a continuous, positive, decreasing function of x for all $x \geq N$ (N a positive integer). Then the series $\sum_{n=N}^{\infty} a_n$ and the integral $\int_N^{\infty} f(x) dx$ both converge or both diverge.

Convergence Tests that we have learned so far:

Test for convergence:

- ① Definition (partial sums, telescoping series)
- ② special series (geometric series w/ $|r| < 1$, p-series w/ $p > 1$)
- ③ Integral test (only for continuous, positive, decreasing "functions")
- ④ Basic comparison test & limit comparison test.

Test for divergence:

- ① Definition (partial sums)
- ② special series (geometric series w/ $|r| \geq 1$, harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$, $\sum_{k=1}^{\infty} (-1)^k$, p-series w/ $p \leq 1$)
- ③ n-th term test ($\lim_{n \rightarrow \infty} a_n \neq 0$)
- ④ Integral test (only for continuous, positive, decreasing "functions")
- ⑤ Basic comparison test & limit comparison test.

p-series $\sum_{k=1}^{\infty} \left(\frac{1}{k^p} \right)$ is called p-series.

when $p > 1$, the p-series converges.

when $p = 1$, this is the harmonic series, therefore diverges.

when $p < 1$, the p-series diverges.

Comparison and Limit Comparison Test

(Basic)

THEOREM 10—Direct Comparison Test

Let $\sum a_n$ and $\sum b_n$ be two series with $0 \leq a_n \leq b_n$ for all n . Then

1. If $\sum b_n$ converges, then $\sum a_n$ also converges.
2. If $\sum a_n$ diverges, then $\sum b_n$ also diverges.

THEOREM 11—Limit Comparison Test

Suppose that $a_n > 0$ and $b_n > 0$ for all $n \geq N$ (N an integer).

1. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ and $c > 0$, then $\sum a_n$ and $\sum b_n$ both converge or both diverge.
2. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ and $\sum b_n$ converges, then $\sum a_n$ converges.
3. If $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \infty$ and $\sum b_n$ diverges, then $\sum a_n$ diverges.

Convergence Tests that we have learned so far:

- Test for convergence:
- ① Definition (partial sums, telescoping series)
 - ② special series (geometric series w/ $|r| < 1$, p-series w/ $p > 1$)
 - ③ Integral test (only for continuous, positive, decreasing "functions")
 - ④ Basic comparison test & limit comparison test.
 - ⑤ Ratio test w/ $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$
 - ⑥ Root test w/ $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$

- Test for divergence:
- ① Definition (partial sums)
 - ② special series (geometric series w/ $|r| \geq 1$, harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$, $\sum_{k=1}^{\infty} (-1)^k$, p-series w/ $p \leq 1$)
 - ③ n-th term test ($\lim_{n \rightarrow \infty} a_n \neq 0$)
 - ④ Integral test (only for continuous, positive, decreasing "functions")
 - ⑤ Basic comparison test & limit comparison test.
 - ⑥ Ratio test w/ $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$
 - ⑦ Root test w/ $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} > 1$

Ratio and Root Tests

DEFINITION A series $\sum a_n$ **converges absolutely** (is **absolutely convergent**) if the corresponding series of absolute values, $\sum |a_n|$, converges.

THEOREM 12—The Absolute Convergence Test

If $\sum_{n=1}^{\infty} |a_n|$ converges, then $\sum_{n=1}^{\infty} a_n$ converges.

$\sum_{n=1}^{\infty} |a_n|$ converges $\Rightarrow \sum_{n=1}^{\infty} a_n$ converges

THEOREM 13—The Ratio Test

Let $\sum a_n$ be any series and suppose that

$$\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \rho.$$

$\Rightarrow$ converges

Then (a) the series **converges absolutely** if $\rho < 1$, (b) the series **diverges** if $\rho > 1$ or ρ is infinite, (c) the test is **inconclusive** if $\rho = 1$.

THEOREM 14—The Root Test

Let $\sum a_n$ be any series and suppose that

$$\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \rho.$$

$\Rightarrow$ converges

Then (a) the series **converges absolutely** if $\rho < 1$, (b) the series **diverges** if $\rho > 1$ or ρ is infinite, (c) the test is **inconclusive** if $\rho = 1$.

Convergence Tests that we have learned so far:

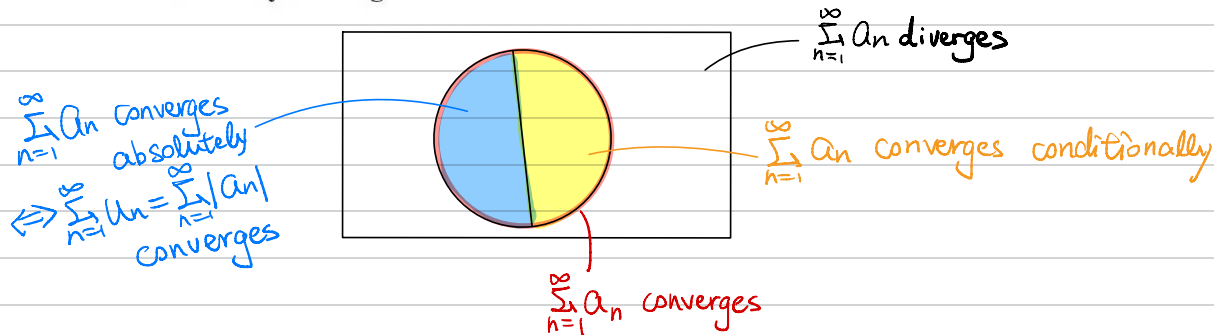
- Test for convergence:
- ① Definition (partial sums, telescoping series)
 - ② special series (geometric series w/ $|r| < 1$, p-series w/ $p > 1$)
 - ③ Integral test (only for continuous, positive, decreasing "functions")
 - ④ Basic comparison test & limit comparison test.
 - ⑤ Ratio test w/ $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$
 - ⑥ Root test w/ $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} < 1$
 - ⑦ Alternating test (for alternating series).

- Test for divergence:
- ① Definition (partial sums)
 - ② special series (geometric series w/ $|r| \geq 1$, harmonic series $\sum_{k=1}^{\infty} \frac{1}{k}$, $\sum_{k=1}^{\infty} (-1)^k$, p-series w/ $p \leq 1$)
 - ③ n-th term test ($\lim_{n \rightarrow \infty} a_n \neq 0$)
 - ④ Integral test (only for continuous, positive, decreasing "functions")
 - ⑤ Basic comparison test & limit comparison test.
 - ⑥ Ratio test w/ $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$
 - ⑦ Root test w/ $\lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} > 1$

Alternating Series

$$\text{Let } u_n = |a_n|$$

DEFINITION A series that is convergent but not absolutely convergent is called **conditionally convergent**.



THEOREM 15—The Alternating Series Test

The series

$$\sum_{n=1}^{\infty} (-1)^{n+1} u_n = u_1 - u_2 + u_3 - u_4 + \cdots$$

converges if the following conditions are satisfied:

1. The u_n 's are all positive.
2. The u_n 's are eventually nonincreasing: $u_n \geq u_{n+1}$ for all $n \geq N$, for some integer N .
3. $u_n \rightarrow 0$.

$$C_0 + C_1(x-a) + C_2(x-a)^2 + C_3(x-a)^3 + \dots$$

Power Series

Standard form: $\sum C_n(x-a)^n$

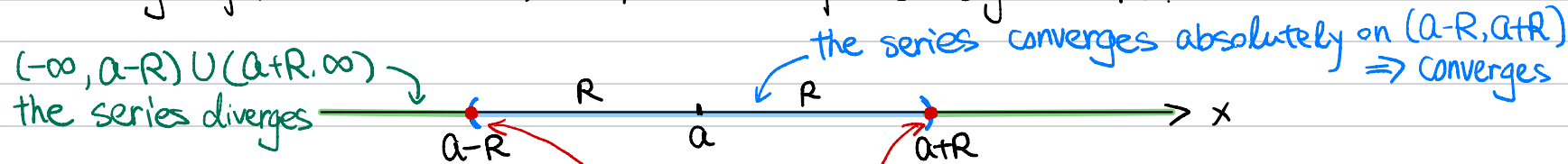
Other forms: $\sum C_n(ax+b)^n$

$\sum C_n(x-a)^n$ is a power series centered at a .

Radius of convergence is a number R s/t the series $\sum C_n(x-a)^n$ diverges when $|x-a| > R$, and the series converges absolutely when $|x-a| < R$. The series may or may not converge at the endpoints, i.e. $|x-a| = R$.

Interval of convergence is the interval for values of x s/t the series converges and the series diverges outside this interval.

e.g. for $\sum C_n(x-a)^n$ w/ radius of convergence R .



$(-\infty, a-R) \cup (a+R, \infty)$
the series diverges

At the two endpoints, the series may or may not converge (conditionally) (You will need to check this separately)

Interval of Convergence:
"[or $(a-R, a+R)$]" or "]"

Taylor Polynomials

DEFINITIONS Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!} (x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!} (x - a)^n + \cdots.$$

DEFINITION Let f be a function with derivatives of order k for $k = 1, 2, \dots, N$ in some interval containing a as an interior point. Then for any integer n from 0 through N , the **Taylor polynomial of order n** generated by f at $x = a$ is the polynomial

$$P_n(x) = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!} (x - a)^2 + \cdots + \frac{f^{(k)}(a)}{k!} (x - a)^k + \cdots + \frac{f^{(n)}(a)}{n!} (x - a)^n.$$

Taylor Polynomials

Taylor's Formula

If f has derivatives of all orders in an open interval I containing a , then for each positive integer n and for each x in I ,

$$\begin{aligned} f(x) = & f(a) + f'(a)(x - a) + \frac{f''(a)}{2!}(x - a)^2 + \cdots \\ & + \frac{f^{(n)}(a)}{n!}(x - a)^n + R_n(x), \end{aligned} \quad (1)$$

where

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - a)^{n+1} \quad \text{for some } c \text{ between } a \text{ and } x. \quad (2)$$

$$\Rightarrow |R_n(x)| \leq \max_{c \in [a, x]} (|f^{(n+1)}(c)|) \frac{(x-a)^{n+1}}{(n+1)!}$$

Taylor Series : Maclaurin Series

DEFINITIONS Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!} (x - a)^k = f(a) + f'(a)(x - a) + \frac{f''(a)}{2!} (x - a)^2 + \cdots + \frac{f^{(n)}(a)}{n!} (x - a)^n + \cdots.$$

The Maclaurin series of f is the Taylor series generated by f at $x = 0$, or

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!} x^k = f(0) + f'(0)x + \frac{f''(0)}{2!} x^2 + \cdots + \frac{f^{(n)}(0)}{n!} x^n + \cdots.$$

e.g. Maclaurin Series for $\sin(x)$:

$$\begin{array}{llll} f'(x) = \cos(x) & f''(x) = -\sin(x) & f'''(x) = -\cos(x) & f^{(4)}(x) = \sin(x) \quad \dots \\ f'(0) = 1 & f''(0) = 0 & f'''(0) = -1 & f^{(4)}(0) = 0 \quad \dots \end{array}$$

$$\begin{aligned} \text{Hence } \sin(x) &= \cancel{f(0)} + f'(0)x + \cancel{\frac{f''(0)}{2!}x^2} + \cancel{\frac{f'''(0)}{3!}x^3} + \cancel{\frac{f^{(4)}(0)}{4!}x^4} + \dots \\ &= \frac{1}{1!}x + \frac{-1}{3!}x^3 + \frac{1}{5!}x^5 + \frac{-1}{7!}x^7 + \dots \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!} \end{aligned}$$

e.g. Maclaurin Series for e^x :

$$\begin{array}{llll} f'(x) = e^x & f''(x) = e^x & f'''(x) = e^x & f^{(4)}(x) = e^x \quad \dots \\ f'(0) = 1 & f''(0) = 1 & f'''(0) = 1 & f^{(4)}(0) = 1 \quad \dots \end{array}$$

$$\begin{aligned} \text{Hence } e^x &= f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \dots \\ &= 1 + 1 \cdot x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \dots \\ &= \sum_{k=0}^{\infty} \frac{x^k}{k!} \end{aligned}$$